

PROJECT REPORT

MIDNIGHT DODGERS

A 2D arcade-style car dodging game developed using C++ and SFML
(Simple and Fast Multimedia Library).

PARTNER

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FRAMEWORK

SFML Version 2.6.2

PROJECT DESCRIPTION

Midnight Dodgers is a 2D arcade-style car dodging game developed using C++ and SFML (Simple and Fast Multimedia Library). The player controls a car on a highway and must avoid incoming enemy vehicles while attempting to survive as long as possible and achieve the highest score.

As the score increases, enemy vehicles move faster, making the game progressively more challenging. The game also includes multiple menus, car selection, high-score tracking, sound effects, and a lives system to enhance the gameplay experience.

LIBRARIES USED

- **SFML (Simple and Fast Multimedia Library)** — Version: 2.6.2

FEATURES

Menu System

The game features a complete menu system including: Play, High Scores, Help, Options, and Exit. The user can navigate through different game screens using mouse clicks and keyboard input.

Player Name Entry

Before starting the game, the player is required to enter their name, which personalizes the gameplay experience.

Car Selection

Players can choose from multiple vehicle designs before entering the game.

Score Counter

The score is displayed during gameplay and increases as the player successfully avoids enemy vehicles.

High Score System

The game stores and displays the highest score achieved using file handling techniques.

Enemy Vehicle Spawning

Enemy vehicles are generated continuously in different lanes and move downward toward the player.

Input Detection

Input detection is implemented using SFML event handling functions. The player can control the car using keyboard inputs.

Player Movement

The player's vehicle can move Left Arrow Key (→ Move Left) and Right Arrow Key (→ Move Right). Movement is restricted within the road boundaries.

Collision Detection

Collision detection is implemented using sprite bounding boxes. When the player's car collides with an enemy vehicle: a life is lost, crash sound effects are played, and safe mode is activated temporarily.

Lives System

The player starts with three lives. The game ends when all lives are exhausted.

Safe Mode

After a collision, the player receives temporary invincibility for a short duration, preventing immediate consecutive collisions.

Dynamic Difficulty

The difficulty gradually increases as the score grows by increasing the speed of enemy vehicles.

Pause and Resume

Players can pause and resume the game at any time using the keyboard.

Audio System

The game includes background music, button click sounds, crash sound effects, and other gameplay audio effects.

Options Menu

Players can adjust volume levels and enable or disable music.

Help Screen

A dedicated help screen provides instructions and controls for the player.

Game Over System

When all lives are lost, a Game Over screen is displayed showing the final score, a restart option, and an exit option.

HOW TO PLAY

- Launch the game.
- Enter your name.
- Select your preferred vehicle.
- Start the game.
- Use the Left and Right Arrow keys to control the car.
- Avoid incoming enemy vehicles.
- Earn points by surviving longer.
- Try to achieve the highest score.
- Use the Pause feature whenever needed.
- The game ends when all three lives are lost.

CONTROLS

KEY	ACTION
Left Arrow	Move Left
Right Arrow	Move Right
P	Pause / Resume
ESC	Exit Game
Mouse Click	Menu Navigation

CONCLUSION

In conclusion, Midnight Dodgers successfully demonstrates the implementation of game development concepts using C++ and SFML. The project integrates graphics rendering, event handling, audio management, file handling, collision detection, and state-based game design into a complete arcade-style game. Through features such as car selection, high-score tracking, multiple game states, and dynamic difficulty adjustment, the game provides an engaging and enjoyable experience for players while showcasing practical programming and software development skills.